

Quant Slides

TGIR QuantLib Tools

TG Investments and Research

2026-04-06

Portfolio Model Scope

- Spot-starting SOFR swap.
- European normal-volatility swaption under Hull-White 1F + Jamshidian.
- Bermudan swaption with calibrated Hull-White 1F or G2++.
- SPX cliquet under Black-Scholes-Merton dynamics.

Rates Curve

- Inputs: workbook SOFR strip from 1D through 30Y.
- Helpers: one deposit plus the workbook OIS helper strip.
- Curve: `ql.PiecewiseLogCubicDiscount`.
- Validation: OIS repricing to near-zero NPV.

Swaption Stack

- European swaption uses one ATM matrix cell, calibrates `ql.HullWhite`, and prices with `ql.JamshidianSwaptionEngine`.
- Bermudan swaption calibrates `ql.HullWhite` or `ql.G2` to the trade call schedule and remaining-tenor strip.
- Pricing uses `ql.TreeSwaptionEngine`.
- Tests now assert helper repricing after calibration and the fixed-maturity call mapping.

Cliquet Stack

- Inputs: SPX spot, dividend yield, flat vol, quantity, moneyness, maturity, reset frequency.
- Process: `ql.BlackScholesMertonProcess`.
- Instrument: `ql.CliquetOption`.
- Engine: `ql.AnalyticCliquetEngine`.

Reported Analytics

- Rates: curve ladder, matrix vega, Bermudan mean-reversion sensitivity.
- Cliquet: analytic Greeks, reset-strip decomposition, deterministic scenarios, Monte Carlo distribution.
- Model page: constructor signatures, enum values, and research lists.

Validation Philosophy

- Reprice the exact calibration instruments wherever possible.
- Use deterministic reduction identities for structured products.
- Keep UI smoke tests and model tests separate.
- Prefer explicit tolerances over snapshot-style golden numbers.